

3.6.2 Domestic electricity

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 3			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Most electrical appliances are connected to the mains using a three-core flex. The insulation covering each wire in the flex is colour-coded for easy identification: • live wire – brown • neutral wire – blue • earth wire – green and yellow stripes. The earth wire is a safety wire to stop the appliance becoming live and the fuse contains a thin piece of wire, which melts if the current becomes too large, thereby cutting off the supply. Students should be able to select the correct fuse from a list when given the current rating of an appliance. Some appliances do not have an earth wire because they are double insulated.		6.2.3.2	4.7.2.6
Outcome 4			
Everyday electrical appliances are designed to bring about energy transfers. The amount of energy an appliance transfers depends on how long the appliance is switched on for and the power of the appliance.	Students may be required to use the equation: $\text{energy (kWh)} = \text{power (kW)} \times \text{time (h)}$	6.4.4.2	4.7.2.8